



S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical 01

Aim: A company wants to understand how employees communicate through email. To achieve this, perform Exploratory Data Analysis (EDA) on an email dataset using Pandas and visualization tools to identify communication patterns, active users, and key observations.

Name of Student:

Roll No.:

Semester/Year:

Academic Session:

Date of Performance:

Date of Submission:

Exploratory Data Analysis Lab (N-PECCD503P)

AIM: A company wants to understand how employees communicate through email. To achieve this, perform Exploratory Data Analysis (EDA) on an email dataset using Pandas and visualization tools to identify communication patterns, active users, and key observations.

IMPLEMENTATION/SCREENSHOTS:

 jupyter Untitled2 Last Checkpoint: 20 seconds ago

File Edit View Run Kernel Settings Help



```
[12]: import pandas as pd
df = pd.read_csv(r"C:\Users\STUDENTS\Desktop\email.csv")
print(df)
```

	email_id	sender	receiver	subject	timestamp
0	1	Alice	Bob	Project Update	1/2/2024 09:15
1	2	Bob	Emma	Meeting Request	1/2/2024 10:20
2	3	Charlie	David	Task Assignment	1/2/2024 11:00
3	4	Emma	Alice	Client Feedback	1/3/2024 09:30
4	5	Frank	Grace	Weekly Report	1/3/2024 10:15
5	6	Henry	Ivy	Budget Review	1/3/2024 11:45
6	7	Jack	Karen	Project Update	1/4/2024 09:05
7	8	Leo	Mia	Meeting Request	1/4/2024 10:40
8	9	Nathan	Olivia	Task Assignment	1/4/2024 13:20
9	10	Peter	Quinn	Weekly Report	1/4/2024 15:10
10	11	Ryan	Sophia	Project Update	1/5/2024 09:00
11	12	Tom	Alice	Client Feedback	1/5/2024 10:10
12	13	Alice	David	Meeting Request	1/5/2024 11:30
13	14	Bob	Frank	Budget Review	1/6/2024 09:45
14	15	Charlie	Henry	Task Assignment	1/6/2024 12:15
15	16	Emma	Jack	Project Update	1/6/2024 14:00
16	17	Grace	Leo	Weekly Report	1/7/2024 09:20
17	18	Ivy	Nathan	Client Feedback	1/7/2024 10:35
18	19	Karen	Peter	Meeting Request	1/7/2024 13:10
19	20	Mia	Ryan	Budget Review	1/7/2024 15:45
20	21	Olivia	Tom	Project Update	1/8/2024 09:05
21	22	Quinn	Bob	Task Assignment	1/8/2024 10:50
22	23	Sophia	Charlie	Weekly Report	1/8/2024 11:25
23	24	Alice	Emma	Project Update	1/9/2024 09:10
24	25	Alice	Bob	Meeting Request	1/9/2024 10:00
25	26	Alice	David	Budget Review	1/9/2024 11:15
26	27	Bob	Alice	Task Assignment	1/10/2024 09:20
27	28	Bob	Emma	Weekly Report	1/10/2024 10:40
28	29	Charlie	Alice	Project Update	1/10/2024 12:00

```

27      28      Bob      Emma      Weekly Report      1/10/2024 10:40
28      29      Charlie Alice      Project Update      1/10/2024 12:00
29      30      David   Alice      Client Feedback     1/10/2024 14:30
30      31      Emma    Bob      Meeting Request      1/11/2024 09:15
31      32      Frank   Alice      Task Assignment      1/11/2024 10:30
32      33      Grace   Bob      Project Update       1/11/2024 11:45
33      34      Henry   Alice      Budget Review        1/11/2024 13:20
34      35      Ivy     Emma      Weekly Report        1/12/2024 09:05
35      36      Jack    Alice      Meeting Request       1/12/2024 10:10
36      37      Karen    Bob      Project Update        1/12/2024 11:55
37      38      Leo     Alice      Task Assignment       1/12/2024 14:10
38      39      Mia     Emma      Client Feedback       1/13/2024 09:35
39      40      Nathan   Bob      Weekly Report         1/13/2024 10:50
40      41      Olivia   Alice      Project Update        1/13/2024 12:05
41      42      Peter    Emma      Meeting Request       1/13/2024 13:30
42      43      Quinn   Alice      Budget Review         1/14/2024 09:15
43      44      Ryan    Bob      Task Assignment       1/14/2024 10:45
44      45      Sophia   Emma      Project Update        1/14/2024 11:20
45      46      Tom     Alice      Weekly Report         1/14/2024 14:00
46      47      Alice    Bob      Project Update        1/15/2024 09:00
47      48      Bob     Alice      Meeting Request       1/15/2024 10:10
48      49      Emma    Alice      Client Feedback       1/15/2024 11:25
49      50      Alice    Emma      Budget Review         1/15/2024 15:00

```

```

[1]: import pandas as pd
      df = pd.read_csv(r"C:\Users\STUDENTS\Desktop\email.csv")
      df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ---
 0   email_id    50 non-null    int64
 1   sender      50 non-null    object
 2   receiver    50 non-null    object
 3   subject     50 non-null    object
 4   timestamp   50 non-null    object
dtypes: int64(1), object(4)
memory usage: 2.1+ KB

```

```

[2]: df.head()

```

```

[2]:
   email_id  sender  receiver  subject  timestamp
0         1    Alice      Bob  Project Update  1/2/2024 09:15
1         2      Bob     Emma  Meeting Request  1/2/2024 10:20
2         3  Charlie   David  Task Assignment  1/2/2024 11:00
3         4    Emma    Alice  Client Feedback  1/3/2024 09:30
4         5    Frank   Grace  Weekly Report   1/3/2024 10:15

```

[3]: `df.tail()`

```
[3]:
```

	email_id	sender	receiver	subject	timestamp
45	46	Tom	Alice	Weekly Report	1/14/2024 14:00
46	47	Alice	Bob	Project Update	1/15/2024 09:00
47	48	Bob	Alice	Meeting Request	1/15/2024 10:10
48	49	Emma	Alice	Client Feedback	1/15/2024 11:25
49	50	Alice	Emma	Budget Review	1/15/2024 15:00

[4]: `df.isnull().sum()`

```
[4]: email_id      0
      sender       0
      receiver    0
      subject     0
      timestamp   0
      dtype: int64
```

[5]: `df.shape`

```
[5]: (50, 5)
```

[9]: `sender_count = df['sender'].value_counts()`

```
print(sender_count)
```

```
sender
Alice    7
Bob      5
```

```
[9]: sender_count = df['sender'].value_counts()
```

```
print(sender_count)
```

```
sender
Alice      7
Bob        5
Emma       4
Charlie    3
Grace      2
Sophia     2
Quinn      2
Olivia     2
Mia        2
Karen      2
Ivy        2
Ryan       2
Tom        2
Peter      2
Nathan     2
Leo        2
Jack       2
Henry      2
Frank      2
David      1
Name: count, dtype: int64
```

```
[10]: receiver_count = df['receiver'].value_counts()
```

```
print(receiver_count)
```

```
receiver
Alice      14
Bob        9
Emma       8
David      3
Henry      1
Tom        1
Ryan       1
Peter      1
Nathan     1
Leo        1
Jack       1
Sophia     1
Frank      1
Quinn      1
Olivia     1
Mia        1
Karen      1
Ivy        1
Grace      1
Charlie    1
Name: count, dtype: int64
```

```
[*]: daily = df['day'].value_counts()
```

```
print(daily)
```

```
print("Rows and Columns:")
print(df.shape)

print("\nDataset Info:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
```

Rows and Columns:
(50, 5)

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
Column Non-Null Count Dtype
--- ----- -----
0 email_id 50 non-null int64
1 sender 50 non-null object
2 receiver 50 non-null object
3 subject 50 non-null object
4 timestamp 50 non-null object
dtypes: int64(1), object(4)
memory usage: 2.1+ KB
None

Missing Values:
email_id 0
sender 0
receiver 0
subject 0
timestamp 0
dtype: int64

```
##### Sender #####
[8]: sender_count = df['sender'].value_counts()
print(sender_count)
plt.figure(figsize=(10,5))

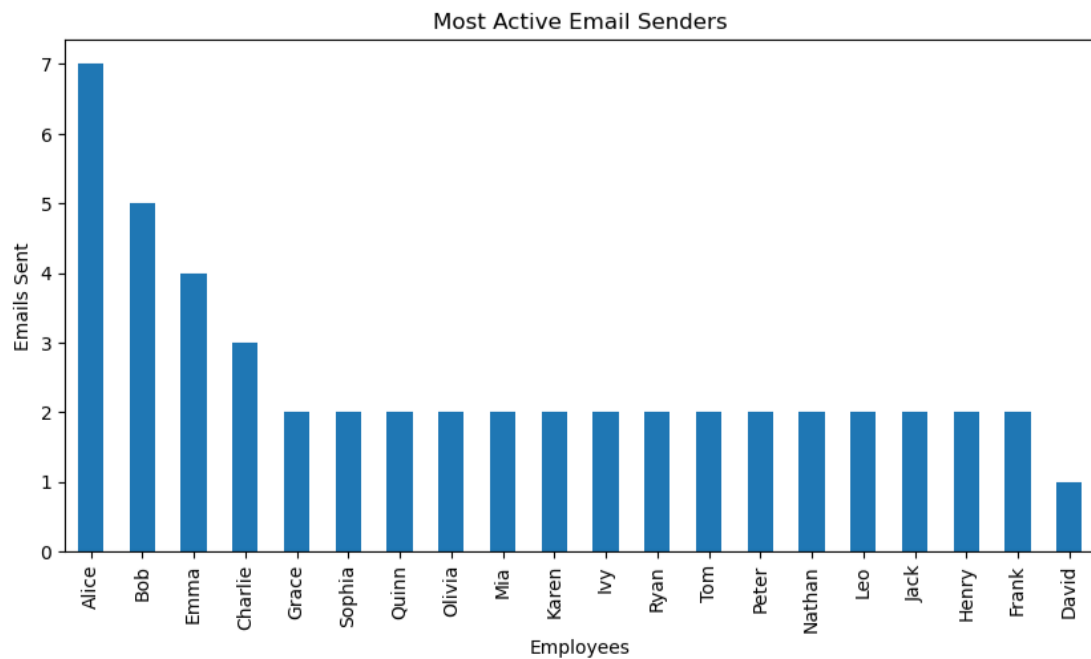
sender_count.plot(kind='bar')

plt.title("Most Active Email Senders")
plt.xlabel("Employees")
plt.ylabel("Emails Sent")

plt.show()
```

```
sender
Alice      7
Bob        5
Emma       4
Charlie    3
Grace      2
Sophia     2
Quinn      2
Olivia     2
Mia        2
Karen      2
Ivy        2
Ryan       2
Tom        2
Peter      2
Nathan     2
Leo        2
Jack       2
Henry      2
Frank      2
David      1
Name: count, dtype: int64
```

```
Frank      2
David      1
Name: count, dtype: int64
```



```
[9]: receiver_count = df['receiver'].value_counts()

print(receiver_count)
plt.figure(figsize=(10,5))

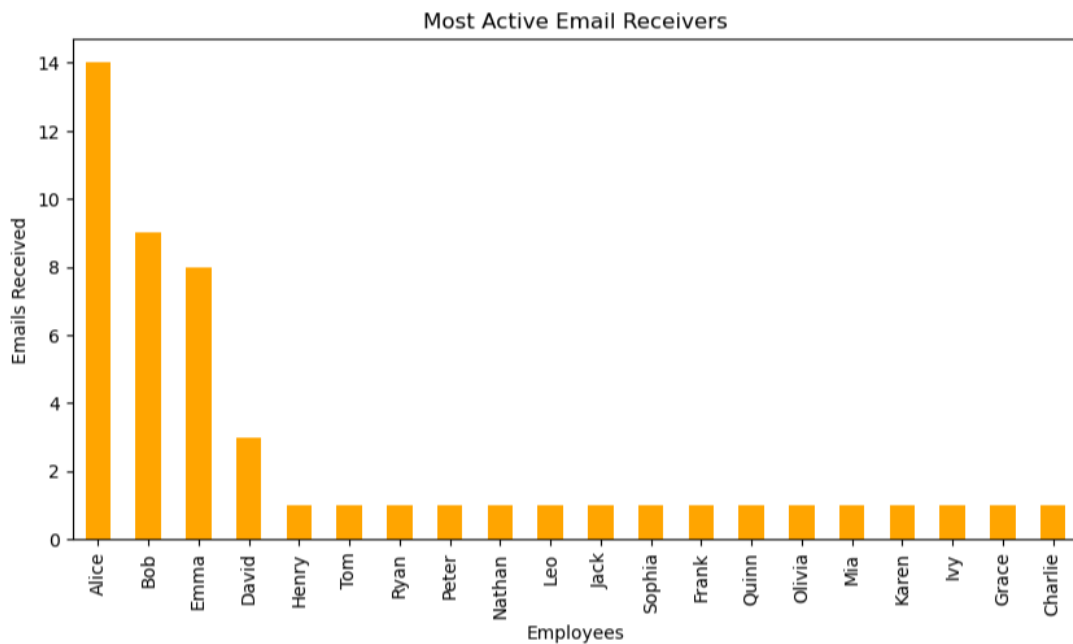
receiver_count.plot(kind='bar', color='orange')

plt.title("Most Active Email Receivers")
plt.xlabel("Employees")
plt.ylabel("Emails Received")

plt.show()
```

```
receiver
Alice      14
Bob         9
Emma        8
David        3
Henry        1
Tom          1
Ryan         1
Peter        1
Nathan       1
Leo          1
Jack         1
Sophia       1
Frank        1
Quinn        1
Olivia       1
Mia          1
Karen        1
Ivy          1
Grace        1
Charlie      1
Name: count, dtype: int64
```

```
Ivy 1
Grace 1
Charlie 1
Name: count, dtype: int64
```



```
[10]: pairs = df.groupby(
        ['sender', 'receiver']
    ).size().reset_index(name='count')

pairs = pairs.sort_values(
    by='count',
    ascending=False
)

print(pairs.head(10))

   sender receiver  count
0    Alice      Bob     3
4     Bob     Emma     2
1    Alice   David     2
10   Emma    Alice     2
41    Tom    Alice     2
2    Alice     Emma     2
3     Bob    Alice     2
5     Bob    Frank     1
33  Peter     Emma     1
27    Mia     Emma     1

[13]: subject_count = df['subject'].value_counts()

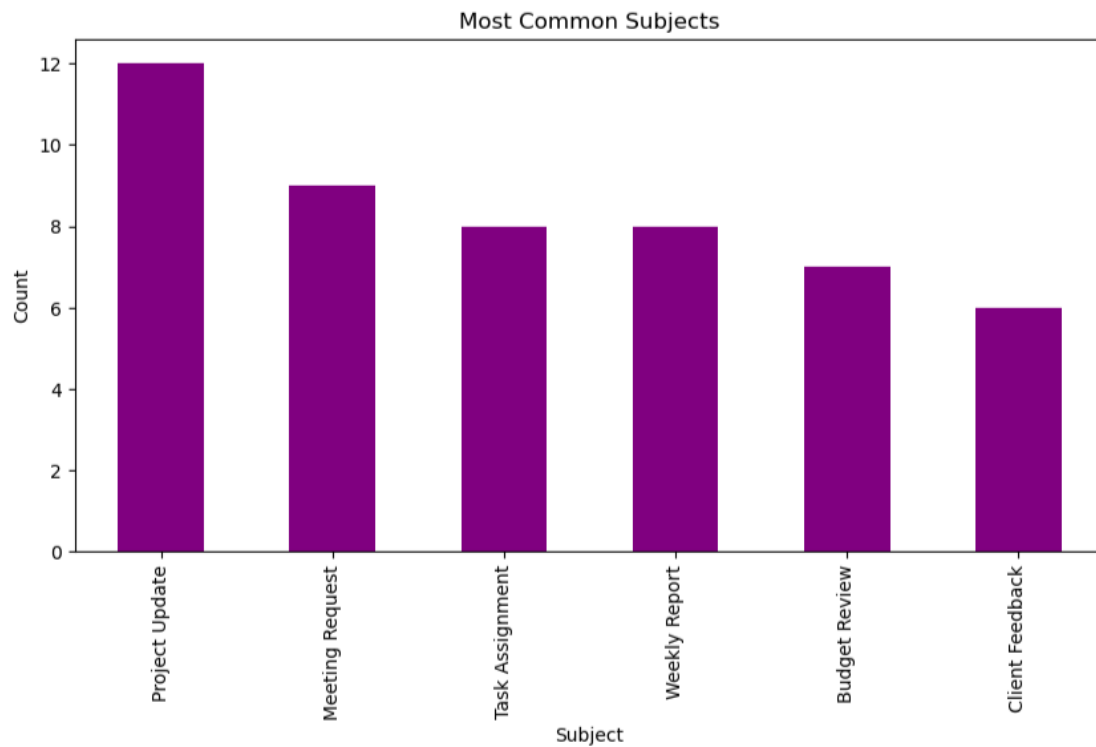
print(subject_count)
plt.figure(figsize=(10,5))

subject_count.plot(
    kind='bar',
    color='purple'
)

plt.title("Most Common Subjects")
plt.xlabel("Subject")
plt.ylabel("Count")

plt.show()
```


Name: count, dtype: int64



```
[15]: print("Most Active Sender:")
print(df['sender'].value_counts().idxmax())

print("\nMost Active Receiver:")
print(df['receiver'].value_counts().idxmax())

print("\nMost Common Subject:")
print(df['subject'].value_counts().idxmax())
```

Most Active Sender:
Alice

Most Active Receiver:
Alice

Most Common Subject:
Project Update

[]: